Multi-level Modeling (MLM) Analysis

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Description: Introduction and tutorial supplement for exploring the utilization of multi-level (hierarchical linear) models to predict a continuous outcome from categorical (dichotomous) condition data nested by participant and featuring repeated, within-subjects' trial measures. This example will focus on predicting *Mean Line Length* (L_{mean}) trial outcomes from cross-recurrence quantification analysis (CRQA) on intrapersonal hand-torso coupling during a cyclical target tracking task under two coordination conditions: Avoid (anti-phase) and Approach (in-phase).

Supplemental Documents: MLM_Tutorial-Corbin.html; MLM_Tutorial-Corbin.rmd

Recommended packages: car, Hmisc, ppcor, psych, MASS, mctest, fBasics, foreign, jtools, lmerTest, lme4, tidyverse, dplyr, ggplot2, data.table, cowplot, performance

Keywords: multi-level modeling, hierarchical linear models, mixed models, growth curve analysis

Introduction

While linear regression and ANOVA analyses offer several flexible options for investigating data trends, making predictions, and uncovering group-dependent differences on outcome variables, these Ordinary Least Squares based approaches provide limited support for *nested data*, or instances in which individual data points cannot be considered independent; the data contains clusters or *hierarchies* that must be considered during data analyses to account for (and differentiate between) individual and group-level effects. In brief, multi-level models avoid flattening the data to group-level means only or disaggregating data into non-independent, individual-level comparisons, instead exploring relationships between *and* across levels in one analysis. Multi-level models are powered by the Maximum Likelihood Estimation (rather than the Ordinary Least Squares of traditional regression approaches) to overcome the interdependence inherent to nested data. Maximum likelihood estimations focus on deriving model parameters based on their likelihood of predicting a particular outcome from a normally distributed dataset; it can be used to estimate the average of your outcome variables, as well as

the variance thereof. In contrast, ordinary least squares approaches focus on minimizing the overall error between a predicted and observed models of the data; it is assumed that variance can be largely captured (and subsequently controlled) by observing the deviations between these values.

Model Assumptions: OLS vs. MLE Methods

Assumptions of MR. Linear regression and multi-level models share a number of common assumptions that determine the efficacy of the analysis on the dataset of interest. For example, both types of analyses assume: 1) *linearity*: ensuring the data are best represented by a linear (rather than a curvilinear, cubic, or otherwise non-linear) model; 2) *no non-zero variance*: predictors should have variance within their values; 3) *no multicollinearity*: the absence of highly intercorrelated predictors avoids redundancy in the variance accounted for by each predictor variable (i.e.: the less the predictors represent the same—or largely overlapping—portions of the model variance, the more likely it is that each predictor will uniquely contribute a portion of explanatory variance in the model outcomes); 4) *homoscedasticity*: assumes that variance of the residuals are uniform across levels of the predictor variable; and 5) *normally-distributed model errors*: assumes that the residuals of the model have a mean of zero are normally distributed; the predictors do not need to be normally distributed as long as the overall model's errors reflect an approximately normal distribution.

Additional Assumptions of MLM. Further, multi-level models additionally assume *normally distributed random intercept* and *slope coefficients* with respect to the overall model. Additionally, violations of *multicollinearity* in multi-level analyses is particularly concerning, as high intercorrelations between predictors can result in less accurate modeling of the fixed effects and/or interactions between nested levels (can be overcome, at least sin part, by *centering* continuous predictors, when applicable; i.e.: subtracting the mean from each case-wise value of the predictor to redistribute the values around a mean of zero). Caveats in the accuracy of resulting MLM models can also reflect limitations due to *sample size* and *power*, so it is especially important to ensure that the data being analyzed sufficiently supports multi-level modeling. Still, working with nested data enables multi-level models to circumvent the linear regression assumptions of *independence of predictors* and *error terms* by considering relations within and across levels of the hierarchy simultaneously; this is done while also effectively handling missing data and non-equal variances between groups (two violations of OLS

assumptions). Acknowledging these assumptions—and more importantly, whether or not the data being analyzed meets them—should result in an analysis that best reflects the data without affecting/biasing the reliability of the model (i.e.: the beta coefficients and model parameters), hence increasing the likelihood that the sample data accurately reflects the population parameters we are interested in observing.

Key Multi-level Modeling Terminology

To fully grasp the utility of multi-level models for nested data analyses, we must define and discuss the following terms: *hierarchical analysis*, *fixed effects*, and *random effects*. *Hierarchical analysis* describes the relationships that exist within and between the levels of nested data within the model; such analyses acknowledge the interdependence of nested data, considering *fixed* and *random* effects at both the individual and group-levels of interpretation. In traditional OLS approaches, random effects are not accessible, as employing them would violate assumptions of the analyses. In hierarchical analyses, including MLMs, added utility to support the nested data structure is offered in the employment of random effects structures.

Fixed effects refer to the regression coefficients that respond to observations/measurements at the group-level of analysis. These sample estimates reflect the differences between groups quite similarly to traditional OLS approaches (i.e.: regression main effects and interaction terms), disregarding the additional context provided in the hierarchy through either aggregation (averaging across groups) or disaggregating (assigning all individuals group-level means as individual data). *Random effects*, on the other hand, allow the regression coefficients to vary across groups, giving the model additional predictive flexibility in the resulting *intercepts* and *slopes*. *Random intercepts* assume that the intercept values for the model vary significantly with respect to individual (or lower level) observations, rather than expecting all individual relations to converge to a single intercept value; employing random intercepts, then, takes some of the onus off the fixed effect to account for such individual differences. Similarly, *random slopes* allow the lower-level observations to vary in relational strength with respect to the association between the predictors and outcome variables; in other words, for some lower-level observations, the effect of the predictor will be stronger/weaker than for others (see *Figure below* for illustrative example of each, reproduced from Fernée et. al., 2021).

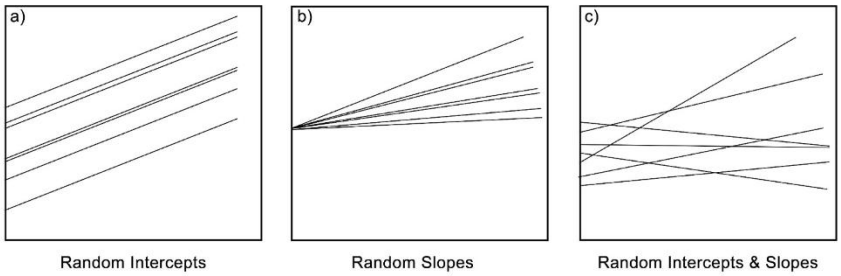


Fig. 3. Types of random effects in multilevel model: a) a random intercepts model, where intercepts the slopes are the same for each group but the intercepts vary, b) A random slope model, where the slopes vary for each group but the intercepts are the same and c) a random intercepts and random slopes model, where both the slopes and intercepts vary by group.

Data Requirements

For this tutorial, a hypothetical dataset was simulated to examine *Mean Line Length* (L_{mean}) trial outcomes from cross-recurrence quantification analysis (CRQA) on intrapersonal hand-torso coupling during a cyclical target tracking task under two coordination conditions: Avoid (anti-phase) and Approach (in-phase). With respect to the ideal final model, the L_{mean} outcomes should reflect a significant effect of the dichotomous condition fixed effect, *movePhase*, while employing a random intercepts and slopes structure with respect to individual participants, *part_ID:*

$$Y_{ij} = (b_0 + u_{0j}) + (b_1 + u_{1j}) * X_{ij} + residual_{error}$$

$$Lmean = (Intercept + randIntEff) + (Slope + randSlopeEff) * movePhase + residual_{error}$$

This corresponds to the following final lmer model in R: `Lmean ~ movePhase + (movePhase|part_ID)`. *Mean line length* is a measure of the predictability/stability of the coordination, with higher values indicating longer average periods of hand-torso coupling. The *movePhase fixed effect* refers to the participant's instructed pointer movement phase during targeting trials; relative to the on-screen target being tracked during each trial, the avoid condition (*movePhase* = 1) involved participants' trackers moving in the opposite direction with respect to the target; the approach condition (*movePhase* = 2) required participants' trackers to move the same direction as the target during trials. Fifty participants were included in this analysis, with a series of within-subjects repeated measures in which each individual completed 6 targeting trials per condition (12 trials per participant), for a grand total of $N_{total.trials}$ = 600. To simulate the data, I chose a grand mean for the dependent variable of 125, employed a large effect size (*Cohen's d* = 2.0, *Pearson's r* = .707), and assigned a variance of 0.30 to calculate the

random intercepts, 0.15 for the random slopes, and a residual variance of 0.05 (see section 2.3 of MLM_Tutorial-Corbin.html for full steps of simulation in R).

Building the Models. In order to analyze our dataset and determine its coherence to our expected final model, we must build 3 comparative models corresponding to the nested relations of interest (see section 3 of MLM_Tutorial-Corbin.html):

- **Base** (random intercept-only): `Lmean ~ (1|part_ID)`
- **Primary** (fixed + random ints effects added): `Lmean ~ movePhase + (1|part_ID)`
- **Random Effects** (fixed + random ints/slopes): `Lmean ~ movePhase + (movePhase|part_ID)`

The Base model focuses on clarifying the necessity of random intercepts; if significant, the random intercepts should be included in the model, indicate that the *intercept* value (unstandardized beta), which represents the predicted value of L_{mean} for all participants' trials before taking into account the condition information (or the grand mean), varies randomly with respect to the participant, with some participants having higher or lower values thereof. As a result, the model should be improved (better predictive precision) by allowing the intercepts to be randomly distributed with respect to *part_ID*. The Primary model seeks to confirm a significant fixed effect of the condition, *movePhase*; significance here, then, corresponds to a significant difference between avoid and approach conditions on L_{mean} outcomes. Finally, the Random Effects model determines if the inclusion of random slopes adds any predictive value to the model; if significant, the random effects model indicates that the *slope* value (unstandardized beta), which represents the strength of the relationship between the *movePhase* condition and resulting value of L_{mean}, is moderated with respect to *part_ID*.

Addressing the Assumptions. Starting with the Base model, values (and visualizations) of skew and kurtosis of the distribution appear to fall within a normal range (value < 2 for $N > 300$). However, the model is met with a violation of two assumptions:

- The residuals vs. fitted values plot indicates a fanning effect (residuals systematically diverge from zero as L_{mean} values increase) indicating a clear violation of *homoscedasticity*.

- The QQ plot indicates that data is insufficiently fitted with respect to a model of only the random intercepts, as indicated by the *non-normal*, s-shaped distribution of the model's plotted values. More specification seems necessary to clarify the ideal model.

For the Primary model:

- Here, the residuals vs. fitted values plot has improved, but we do still see a spreading of the error variance at the extremes of the data. The addition of the *movePhase* fixed effect seems to improve the model, but not completely.
- The QQ plot has also improved, but we do still see some deviation from normal for extreme values. More specification might offer further benefits (i.e.: the random intercepts + slopes model).

And for the assumptions of the Random Effects model:

- The residuals vs. fitted values plot shows marked improvement, with most of the residuals being evenly and randomly distributed around zero (note the general coherence to a horizontal band around the 0 line). Including the random effects appears to further improve the initial (Base model) violations of *homoscedasticity*.
- The QQ plot also shows further improvements, with most points fitting along the line of normality (or within the 95% confidence interval).

Reviewing the assumptions seems to support the employment of the Random effects model (which is promising considering the data simulation requirements), but it remains to be determined if these additional models *actually* ***statistically*** *significant* and which model is the ***ideal***?

Evaluating and Comparing the MLM Models

Now that our models are built and the assumptions have been reviewed, we can evaluate the statistical significance of the three models (see section 3 of MLM_Tutorial-Corbin.html). Beginning with the base model:

- Regarding the Base model, in which L_{mean} is predicted by only the random intercepts, we see a significant *part_ID* random effect, $AIC = 1{,}620.98$, $p < .001$.
- Varying intercepts indicate that the *intercept* value for participants' trials before taking into account the condition information varies randomly with respect to the participant, with some participants having higher or lower values thereof. As a result, the model is

improved (i.e.: offers better predictive precision) by allowing the intercepts to be randomly distributed with respect to *part_ID*.

Still, for the reasons addressed during assumptions, and to investigate the fixed effect of *movePhase*, we will continue with the Primary model:

- Regarding the Primary model, in which L_{mean} is predicted by a fixed effect of *movePhase* and a random intercepts effect, there was a significant effect of *movePhase*, $b = 0.73$ ($SE = .02$), $t(549) = 38.19$, $p < .001$. This indicates that, for those in the Avoid condition, L_{mean} values were significantly lower than those in the Approach condition.

Finally, to address the utility of the addition of random slopes to accompany the random intercepts, we reflect on the Random Effects model:

- Regarding the Random effects model, in which L_{mean} is predicted by a fixed effect of *movePhase* and random slope and intercept effects of *part_ID*, there was a significant effect of adding the random *part_ID* slope effect on *movePhase* (in addition to the random *part_ID* intercept), $AIC = 226.10$, $p < .001$.
- Further, varying slopes indicate that the fixed effect of *movePhase* is moderated with respect to *part_ID*, with the condition having stronger effects on L_{mean} values for some, and weaker effects for others. This is in addition to the *part_ID* random intercept effect.

Great! It appears that our data successfully supports the employment of a random effects structure. Still, we can check this by submitting the Random Effects model to a *stepwise* lmer regression, in which the data is atheoretically fit to the best model. Loading in the Random Effects model (which contains all fixed and random effects of interest) into the step model test revealed results that correspond with those of the MLM analysis, in which the optimal model is that of the slopes model, **Random Effects** `Lmean ~ movePhase + (movePhase|part_ID).`

In sum: R (R core team, 2014) and lmer (Bates, 2005) to perform a multi-level model to test the association between *movePhase* and L_{mean} outcomes during an intrapersonal targeting task. Movement phase condition (*movePhase:* Avoid vs. Approach) was included as a fixed effect, with random intercept and slope effects by participant (*part_ID*). As a result, there was a significant effect of *movePhase*, $b = 0.73$ ($SE = .02$), $t(549) = 38.19$, $p < .001$. This indicates that, for those in the Avoid condition, L_{mean} values were significantly lower than

those in the Approach condition. This result is considered with respect to a significant random effects structure (intercepts and slopes).

Additional Resources

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